

Section 1

Task 1 Latent variable. We introduce a latent allocation variable $z_n \in \{1, \dots, K\}$ indicating which mixture component generated image x_n . Using a binary representation,

$$z_{nk} \in \{0, 1\}, \quad \sum_{k=1}^K z_{nk} = 1,$$

so $z_{nk} = 1$ means that observation x_n is assigned to component k .

The model parameters are:

1. Mixture weights π_k with $\sum_{k=1}^K \pi_k = 1$,
2. Bernoulli pixel probabilities $\mu_{kd} = P(x_{nd} = 1 \mid z_n = k)$.

Complete-data log-likelihood.

Under the Bernoulli product mixture model, the complete-data log-likelihood is

$$\log p(X, Z \mid \pi, \mu) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} \left[\log \pi_k + \sum_{d=1}^D \left(x_{nd} \log \mu_{kd} + (1 - x_{nd}) \log(1 - \mu_{kd}) \right) \right].$$

Expectation-Maximization.

EM maximises the observed likelihood by iteratively maximising

$$Q(\pi, \mu) = \mathbb{E}_{Z \mid X, \pi^{old}, \mu^{old}} [\log p(X, Z \mid \pi, \mu)].$$

E-step. Compute the responsibilities $\gamma_{nk} = P(z_n = k \mid x_n, \pi^{old}, \mu^{old}) = \frac{\pi_k^{old} p(x_n \mid \mu_k^{old})}{\sum_{j=1}^K \pi_j^{old} p(x_n \mid \mu_j^{old})}$,

M-step. Let $N_k = \sum_{n=1}^N \gamma_{nk}$. Maximising Q gives the updates

$$\pi_k = \frac{N_k}{N}, \quad \mu_{kd} = \frac{\sum_{n=1}^N \gamma_{nk} x_{nd}}{N_k}.$$

Those two updating rules obtained by taking partial derivative against Q .

Task 2 In the E-step the likelihood $p(x_n \mid \mu_k) = \prod_d p(x_{nd} \mid \mu_{kd})$ is a product of many numbers in $(0, 1)$. For large dimension (e.g. $D = 784$), this product becomes extremely small (e.g. $0.5^{784} \approx 10^{-236}$) and underflows to zero, making the posterior normalisation undefined.

Therefore the implementation works in the log domain: $a_k = \log \pi_k + \log p(x_n \mid \mu_k)$, turning products into sums. The responsibilities are computed via $\gamma_{nk} = \frac{e^{a_k - m}}{\sum_j e^{a_j - m}}$, $m = \max_j a_j$. Which is mathematically identical to the original posterior but numerically stable (log-sum-exp). Thus the correct probabilities are preserved while avoiding floating-point underflow.

Task 3 Effect of N , EM iterations and the initialisation of μ .

The total negative log-likelihood increases with N because the log-likelihood sums over all observations and therefore scales linearly with the dataset size. After normalising by N , the average NLL approaches a limiting value rather than decreasing. This quantity estimates $\mathbb{E}_{p_{\text{data}}} [-\log p_{\theta}(x)]$, the cross-entropy between the data distribution and the model. Since the Bernoulli mixture assumes conditional independence between pixels, it cannot capture the spatial correlations of handwritten digits, and the loss plateaus as N increases. Different initialisation ranges for μ were also tested. For reasonable ranges (e.g. $[0.1, 0.3]$, $[0.4, 0.6]$, $[0.49, 0.51]$), the final normalised losses were very similar, indicating convergence to solutions with comparable likelihood values, while extreme initialisations performed slightly worse. The EM iterations monotonically decrease the NLL and the improvement becomes small after several iterations, indicating convergence of the algorithm.

Choice of the number of components K . The number of components K controls the capacity of the mixture model. Increasing K always reduces the training NLL because mixture models form a nested family and additional components can be assigned negligible weights. Hence the likelihood improvement for large K is expected.

When K is smaller than the number of digits, a single component must explain multiple digit types, leading to poor fit. For $K = 3$ (the number of distinct digits in the dataset), each component becomes interpretable and captures one digit pattern. For larger K , the likelihood still improves but the extra components mainly model intra-class variability (e.g. different writing styles) rather than new semantic classes. Thus mixture components represent modes of the data distribution rather than class labels.

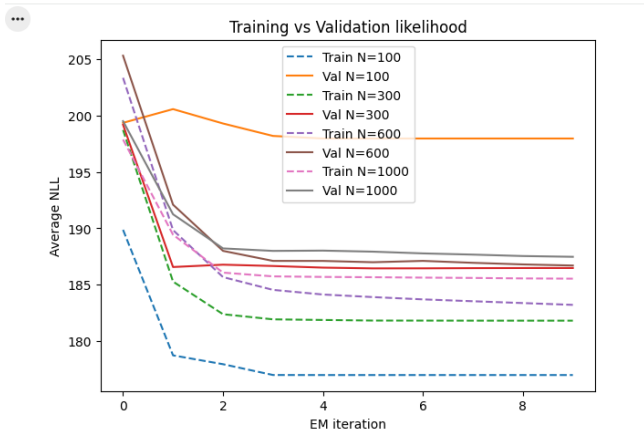


Figure 1: The loss against different sample size N

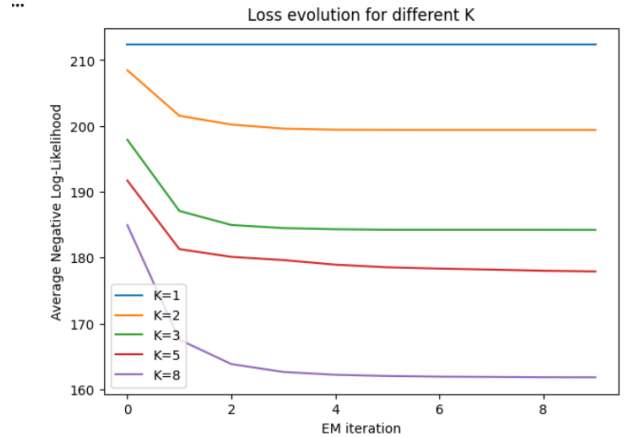


Figure 2: The NLL against different K

Task 4 Task 4: Empirical convergence of EM. We examined the empirical convergence of the EM algorithm by running the updates multiple times with identical hyperparameters but different initial parameter values. The training negative log-likelihood decreases at every iteration for all runs, reflecting the monotonic increase of the data log-likelihood under EM. The decrease is rapid during the first few iterations and then stabilises, indicating convergence in practice.

However, the final likelihood values differ across initialisations. Although each run converges, they converge to different local optima. This shows that the likelihood surface of the mixture model is non-convex and the EM algorithm is sensitive to the initial conditions. Therefore EM guarantees monotonic improvement of the likelihood but not convergence to the global optimum, and the final estimate depends on the starting parameters.

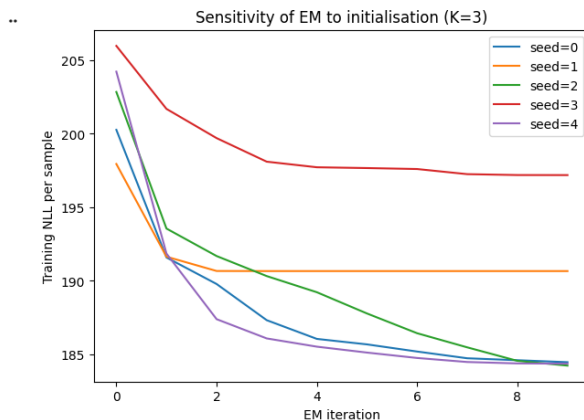


Figure 3: Convergence monotonicity of EM step, and sensitivity towards initial condition

Task 5 Note: I choose the digit to be $[0,3,8]$ in this coursework Each Bernoulli mean μ_k can be reshaped into a 28×28 image, where

$$\mu_{kd} = P(x_d = 1 \mid z = k)$$

represents the probability that pixel d is active given component k . Thus, μ_k is not a single digit image but a probability template.

The visualisations resemble blurred handwritten digits, with the components corresponding approximately to distinct digit patterns. Bright regions indicate pixels frequently activated under that component, while darker regions correspond to rarely active pixels. Hence each component describes a typical mode of the data distribution rather than a deterministic prototype image.

The smooth appearance arises from the Bernoulli independence assumption: the model records how often each pixel is activated but does not model spatial correlations between pixels. Consequently, the mixture model captures distributional structure of the digits rather than exact images.

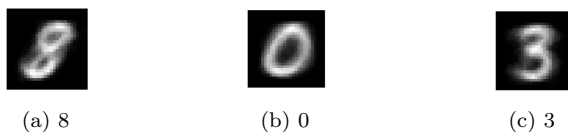


Figure 4: Examples of generated digit images

Task 6 To generate synthetic images, we sample a component index $z \sim \text{Categorical}(\pi)$ and then draw pixels independently as

$$x_d \sim \text{Bernoulli}(\mu_{zd}), \quad d = 1, \dots, D.$$

The sampled images are noisy and do not resemble clean handwritten digits. This follows from the model assumption $p(x \mid z) = \prod_d p(x_d \mid z)$, which treats pixels as conditionally independent. In real handwritten digits, neighbouring pixels are strongly correlated and form continuous strokes, but independent sampling breaks these spatial dependencies and produces scattered patterns. Since μ_{zd} represents activation probabilities rather than deterministic pixel values, sampling generates random realisations instead of smooth digit shapes. Hence the poor visual quality reflects the independence assumption of the Bernoulli mixture model rather than a failure of training.

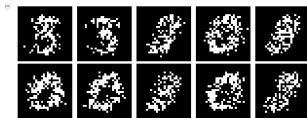


Figure 5: Sampled images from the learnt model

Task 7 The modified Bernoulli parameters push probabilities away from 0.5. For $\alpha = 1$ (original model), many pixel probabilities are close to 0.5, so sampling behaves like independent coin flips and produces noisy images. Increasing α (e.g. $\alpha = 8$) moves probabilities closer to 0 or 1, making the Bernoulli variables nearly deterministic and reducing sampling variance. The generated digits therefore appear clearer and more recognizable.

However, this does not improve the learnt model itself: the independence assumption remains unchanged and spatial correlations are still not modelled. The improvement arises from sharpening the sampling distribution (lower entropy) rather than better representation of digit structure.



Figure 6: Sampled images after sharpening the parameter μ

Section 2

Task 1 I implemented the missing noiseless covariance functions for the SE and Laplace kernels (used throughout Tasks 2–7), with hyperparameters packed in θ .

Task 2 Effect of the input range and comparison with two kernels:

Figure 11 shows prior samples generated with fixed hyperparameters $\theta = (l, \sigma^2, \sigma_\epsilon^2) = (5, 25, 1)$ while only changing the input range. The behaviour of the sampled functions is governed by the ratio between the domain size and the length-scale l .

When input distances are of the same order as l , neighbouring points remain strongly correlated because the kernel covariance $k(x, x')$ is still large, and the sampled trajectories vary smoothly across the domain. As the range becomes much larger than the length-scale ($|x - x'| \gg l$), the covariance rapidly decays and function values at different locations become nearly independent, producing high-frequency noise-like trajectories.

This effect appears for both kernels, but the Laplace kernel yields visibly rougher paths than the SE kernel, since its covariance decreases more rapidly with distance. Hence the length-scale should be interpreted as a correlation distance: smooth functions arise only when the input range is comparable to l .

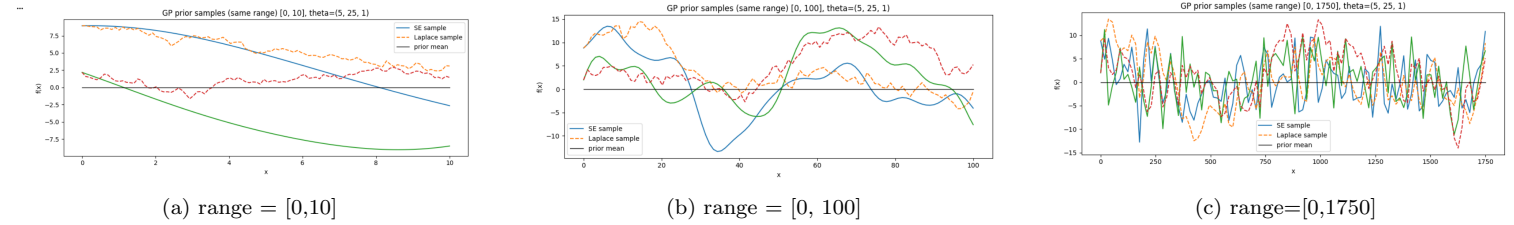


Figure 7: Effects to the trajectories from GP prior by change range space

Effect of other hyperparameters l and σ^2 : Figure 11 illustrates how the GP prior trajectories change when varying the hyperparameters while keeping the input range fixed. The length-scale l controls the smoothness of the sampled functions: smaller values of l reduce the correlation between neighbouring inputs and produce rapidly varying, irregular trajectories, whereas larger l enforces stronger local correlation and yields smoother curves.

In contrast, the marginal variance σ^2 affects the vertical scale of the function without altering its qualitative shape. Since $k(x, x) = \sigma^2$, increasing σ^2 widens the prior credible interval and amplifies the magnitude of the trajectories, while smaller values concentrate the samples closer to the prior mean. Hence l determines how fast the function changes across the input space, whereas σ^2 determines how far it deviates from the mean.

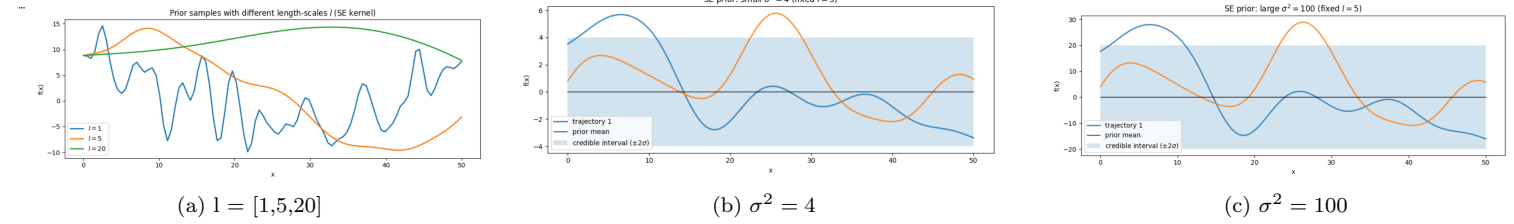


Figure 8: Effects to the trajectories from GP prior by change other hyperparameters

Task 3 The negative log marginal likelihood was implemented following Algorithm 2 in the lecture notes, using a Cholesky-based evaluation of the marginal likelihood for numerical stability.

Task 4 Effect of initial hyperparameters

GP training was repeated with different initialisations of $\theta = (l, \sigma^2, \sigma_\epsilon^2)$. Most initialisations converged to similar low NLL values, indicating stable optimisation. However, some extreme initialisations produced a much larger NLL and nearly identical results for the two kernels.

This occurs when the learned length-scale becomes very small relative to the input spacing. The covariance between distinct points then vanishes and the kernel matrix approaches

$$K \approx (\sigma^2 + \sigma_\epsilon^2)I,$$

so the GP explains the data as independent noise rather than a smooth function. In this regime both kernels reduce to the same diagonal covariance, and the optimiser converges to a noise-dominated solution (a degenerate basin of the marginal likelihood) rather than the desired functional solution.

	init_theta	SE_NLL	Laplace_NLL	better						
0	[0.5, 1.0, 0.01]	418.875931	431.820267	SE						
1	[1.0, 4.0, 0.1]	418.875931	431.820267	SE						
2	[5.0, 25.0, 1.0]	418.875931	431.820267	SE						
3	[20.0, 100.0, 5.0]	418.875931	431.820267	SE						
4	[50.0, 25.0, 1.0]	418.875931	431.820267	SE						
5	[0.1, 25, 1]	558.526829	558.526829	Laplace						
6	[0.0001, 25, 1]	558.526829	558.526829	Laplace						

					fraction	num_points	SE_NLL	Laplace_NLL	better	
					0	0.05	90	226.493386	239.578927	SE
					1	0.10	180	418.875931	431.820267	SE
					2	0.20	360	743.826066	749.608871	SE
					3	0.40	720	1295.002055	1246.533273	Laplace
					4	0.80	1440	2292.277323	2126.277317	Laplace

(a) Initial condition change

(b) Training points size change

Figure 9: Change of NLL via different hyperparameters choice

Changing number of training points: The experiment was repeated using increasing fractions of the dataset. For small training sets (5%–20%), the SE kernel achieved a lower NLL than the Laplace kernel. With limited data, the strong smoothness assumption of the SE kernel acts as a regularising prior and avoids fitting small fluctuations.

As the number of observations increases (40% and 80%), the Laplace kernel becomes preferred. With more data, the marginal likelihood places greater importance on data fit relative to the complexity penalty $\frac{1}{2} \log |K|$, and the rougher covariance structure of the Laplace kernel better explains local variations in the observations.

This illustrates Bayesian model selection via marginal likelihood: with little data smoother models are favoured, while with more data the evidence supports more flexible covariance structures. accurately reflects the true structure of the signal is selected.

Optimizer choice:Here I use the same value of the initial condition as the first part of this task. But I change the optimizer to 'Powell'. To examine the effect of the optimisation algorithm, the training was repeated with a different optimiser. Two types of outcomes were observed: in some cases the optimiser converged to a low NLL (about 418–431), while in others it reached much larger values (around 516–558). The larger NLL values produced nearly identical results for both kernels, indicating convergence to a poor local optimum where the data are effectively treated as noise. When the optimiser reached the lower NLL solution, the results were consistent and allowed reliable model comparison. Different optimisers exhibit different sensitivity to initialisation because they explore the non-convex marginal likelihood landscape along different search trajectories, and may therefore converge to different attraction basins.

	init_theta	SE_NLL	Laplace_NLL	better
0	[0.5, 1.0, 0.01]	418.875931	431.821235	SE
1	[1.0, 4.0, 0.1]	418.876500	447.580864	SE
2	[5.0, 25.0, 1.0]	418.876058	431.886782	SE
3	[20.0, 100.0, 5.0]	418.876011	437.182561	SE
4	[50.0, 25.0, 1.0]	418.877282	558.526829	SE
5	[0.1, 25, 1]	418.876168	431.944419	SE
6	[0.0001, 25, 1]	418.876168	431.944588	SE

Figure 10: Effects of optimizer

Task 5 We implemented the Gaussian Process posterior inference. Given the training observations, the posterior mean and covariance were computed using the conditional Gaussian equations. Posterior trajectories were then sampled from the resulting multivariate normal distribution via a Cholesky decomposition, and stored for later visualisation.

Task 6. Both kernels fit the observations well in regions with dense data, where the posterior mean closely follows the measurements and the uncertainty band becomes narrow.

However, clear qualitative differences appear between the kernels. The SE kernel produces a much smoother posterior function, with gently varying oscillations and highly correlated neighbouring points. In contrast, the Laplace kernel yields a rougher posterior with sharper local variations and more irregular trajectories.

Moreover, the uncertainty away from the observations grows more rapidly for the Laplace kernel, reflecting its shorter correlation length and stronger locality assumption. Overall, the SE kernel imposes a strong smoothness prior, while the Laplace kernel allows more local flexibility in the reconstructed signal.

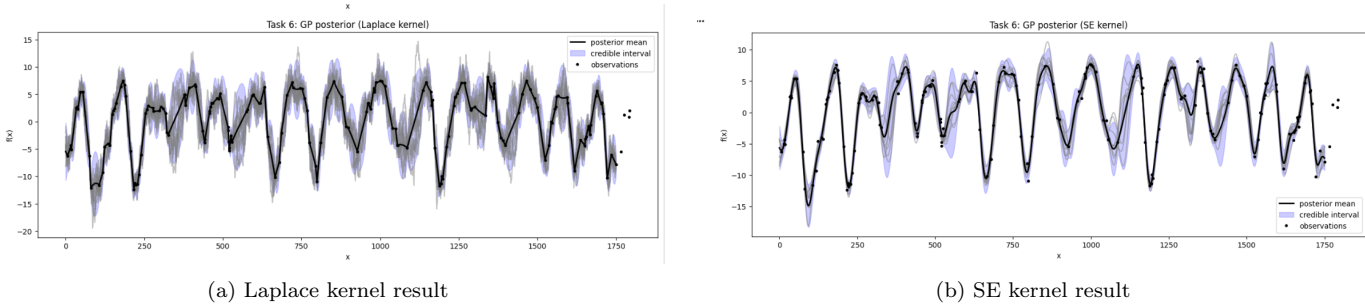


Figure 11: Example of posterior with different kernel

Task 7. The effect of training sample size was evaluated using the posterior mean MSE on the full dataset. For small training sets (10%–20%), both kernels perform similarly and the SE kernel is slightly preferable, as its strong smoothness assumption provides regularisation when data are scarce.

As the number of observations increases (40%–60%), the Laplace kernel achieves significantly lower MSE. With more data, the posterior is increasingly governed by the likelihood rather than the prior, and the more flexible covariance structure of the Laplace kernel better captures local variations in the signal.

This demonstrates a bias–variance trade-off: smooth kernels are favoured with limited data, while more flexible kernels are preferred when sufficient observations are available.

Train %	NLL		MSE	
	SE	Laplace	SE	Laplace
10%	440.489343	441.950663	4.723286	4.179538
20%	789.895969	788.376911	1.724418	1.356126
40%	1,363.576598	1,320.850390	0.918927	0.510178
60%	1,863.229704	1,762.289241	0.730395	0.265459

Figure 12: Quantitive assessment with different sample size

Section 3

Variational family

Consider a latent variable model with joint density $p(x, z, \theta)$. We use the variational family

$$q(z, \theta) = q(z)q_\theta(\theta), \quad q_\theta(\theta) = \delta(\theta - \theta_0).$$

The evidence lower bound (ELBO) is

$$\mathcal{L}(q) = \mathbb{E}_{q(z, \theta)} [\log p(x, z, \theta) - \log q(z, \theta)].$$

Substituting $q(z, \theta) = q(z)\delta(\theta - \theta_0)$,

$$\mathcal{L}(q, \theta_0) = \mathbb{E}_{q(z)\delta(\theta - \theta_0)} [\log p(x, z, \theta)] - \mathbb{E}_{q(z)\delta(\theta - \theta_0)} [\log q(z) + \log \delta(\theta - \theta_0)].$$

Using the identity

$$\mathbb{E}_{q(z)\delta(\theta - \theta_0)} [f(z, \theta)] = \mathbb{E}_{q(z)} [f(z, \theta_0)],$$

we obtain

$$\mathcal{L}(q, \theta_0) = \mathbb{E}_{q(z)} [\log p(x, z, \theta_0)] - \mathbb{E}_{q(z)} [\log q(z)] + \text{const.}$$

We use the fact $p(x, z, \theta) = p(x, z|\theta)p(\theta)$ Hence

$$\mathcal{L}(q, \theta_0) = \mathbb{E}_{q(z)} [\log p(x, z|\theta_0)] - \mathbb{E}_{q(z)} [\log q(z)] + \log p(\theta_0) + \text{const.}$$

E-step

Fix θ_0 and optimise the ELBO with respect to $q(z)$:

$$\mathcal{L}(q) = \mathbb{E}_{q(z)} [\log p(x, z|\theta_0)] - \mathbb{E}_{q(z)} [\log q(z)] + \text{const.}$$

Using

$$p(z|x, \theta_0) = \frac{p(x, z|\theta_0)}{p(x|\theta_0)},$$

we write

$$\begin{aligned} \log p(x, z|\theta_0) &= \log p(z|x, \theta_0) + \log p(x|\theta_0). \\ \mathcal{L}(q) &= \mathbb{E}_{q(z)} [\log p(z|x, \theta_0)] - \mathbb{E}_{q(z)} [\log q(z)] + \log p(x|\theta_0). \end{aligned}$$

The first two terms form a Kullback–Leibler divergence:

$$\mathcal{L}(q) = -\text{KL}(q(z) \parallel p(z|x, \theta_0)) + \log p(x|\theta_0).$$

Since $\text{KL}(\cdot \parallel \cdot) \geq 0$, the ELBO is maximised when

$$q^{\text{new}}(z) = p(z|x, \theta_0^{\text{old}}).$$

This is exactly the E-step of the EM algorithm.

M-step

Now fix $q(z)$ and optimise with respect to θ_0 . The ELBO terms depending on θ_0 are

$$\mathcal{L}(\theta_0) = \mathbb{E}_{q(z)} [\log p(x, z|\theta_0)] + \log p(\theta_0) + \text{const.}$$

Therefore

$$\theta_0^{\text{new}} = \arg \max_{\theta_0} \left(\mathbb{E}_{q(z)} [\log p(x, z|\theta_0)] + \log p(\theta_0) \right).$$

Conclusion

Under the restricted variational family

$$q(z, \theta) = q(z)\delta(\theta - \theta_0),$$

coordinate ascent variational inference produces:

- Optimisation of $q(z)$: $q(z) = p(z|x, \theta)$ (E-step)
- Optimisation of θ_0 : maximisation of expected complete-data log-likelihood (M-step)

Hence the Expectation–Maximisation algorithm is a special case of variational inference in which the variational posterior over parameters is constrained to a point mass.

$$\text{EM is a special case of variational inference.}$$

Section 4

Limit of the α -divergence as $\alpha \rightarrow 1$. Define

$$I(\alpha) = \int p(x)^{\frac{1-\alpha}{2}} q(x)^{\frac{1+\alpha}{2}} dx.$$

Then

$$D_\alpha(p\|q) = 4 \frac{1 - I(\alpha)}{1 - \alpha^2}.$$

Step 1: Identification of the indeterminate form.

Evaluating $I(\alpha)$ at $\alpha = 1$,

$$I(1) = \int p(x)^0 q(x)^1 dx = \int q(x) dx = 1.$$

Hence, as $\alpha \rightarrow 1$,

$$1 - I(\alpha) \rightarrow 0, \quad 1 - \alpha^2 \rightarrow 0,$$

and therefore the limit of $D_\alpha(p\|q)$ is of type 0/0.

Step 2: Differentiability of $I(\alpha)$.

Rewrite $I(\alpha)$ as

$$I(\alpha) = \int q(x) \left(\frac{p(x)}{q(x)} \right)^{\frac{1-\alpha}{2}} dx = \int q(x) \exp\left(\frac{1-\alpha}{2} \log \frac{p(x)}{q(x)} \right) dx.$$

Since p and q are mutually absolutely continuous, the ratio $\frac{p(x)}{q(x)}$ and $\log \frac{p(x)}{q(x)}$ are well-defined almost everywhere. Moreover, under standard integrability conditions, the derivative with respect to α can be interchanged with the integral by using dominated convergence theorem. Therefore $I(\alpha)$ is differentiable near $\alpha = 1$ and

$$I'(\alpha) = \int \frac{\partial}{\partial \alpha} \left[q(x) \exp\left(\frac{1-\alpha}{2} \log \frac{p(x)}{q(x)} \right) \right] dx.$$

Applying the chain rule,

$$\frac{\partial}{\partial \alpha} \left[q(x) \exp\left(\frac{1-\alpha}{2} \log \frac{p}{q} \right) \right] = q(x) \exp\left(\frac{1-\alpha}{2} \log \frac{p}{q} \right) \left(-\frac{1}{2} \log \frac{p(x)}{q(x)} \right).$$

Hence,

$$I'(\alpha) = -\frac{1}{2} \int q(x) \left(\frac{p(x)}{q(x)} \right)^{\frac{1-\alpha}{2}} \log \frac{p(x)}{q(x)} dx.$$

Evaluating at $\alpha = 1$,

$$\left(\frac{p}{q} \right)^{\frac{1-\alpha}{2}} = 1,$$

so

$$I'(1) = -\frac{1}{2} \int q(x) \log \frac{p(x)}{q(x)} dx = \frac{1}{2} \int q(x) \log \frac{q(x)}{p(x)} dx = \frac{1}{2} \text{KL}(q\|p).$$

Step 3: Application of L'Hôpital's rule.

Since both numerator and denominator tend to zero and are differentiable near $\alpha = 1$, L'Hôpital's rule applies:

$$\lim_{\alpha \rightarrow 1} D_\alpha(p\|q) = 4 \lim_{\alpha \rightarrow 1} \frac{1 - I(\alpha)}{1 - \alpha^2} = 4 \lim_{\alpha \rightarrow 1} \frac{-I'(\alpha)}{-2\alpha} = 4 \lim_{\alpha \rightarrow 1} \frac{I'(\alpha)}{2\alpha}.$$

Thus,

$$\lim_{\alpha \rightarrow 1} D_\alpha(p\|q) = 4 \cdot \frac{I'(1)}{2} = 2I'(1) = 2 \cdot \frac{1}{2} \text{KL}(q\|p) = \text{KL}(q\|p).$$

$$\boxed{\lim_{\alpha \rightarrow 1} D_\alpha(p\|q) = \text{KL}(q\|p)}$$

Limit as $\alpha \rightarrow -1$ (sketch by symmetry). Factor out the $p(x)$ term:

$$p(x)^{\frac{1-\alpha}{2}} q(x)^{\frac{1+\alpha}{2}} = p(x) p(x)^{-\frac{1+\alpha}{2}} q(x)^{\frac{1+\alpha}{2}} = p(x) \left(\frac{q(x)}{p(x)} \right)^{\frac{1+\alpha}{2}}.$$

Hence,

$$I(\alpha) = \int p(x) \left(\frac{q(x)}{p(x)} \right)^{\frac{1+\alpha}{2}} dx = \int p(x) \exp\left(\frac{1+\alpha}{2} \log \frac{q(x)}{p(x)} \right) dx.$$

As $\alpha \rightarrow -1$, we have $I(-1) = \int p(x) dx = 1$, so $D_\alpha(p\|q)$ again yields a 0/0 indeterminate form. Differentiating under the integral sign and applying L'Hôpital's rule proceeds exactly as in the $\alpha \rightarrow 1$ case, leading to

$$\lim_{\alpha \rightarrow -1} D_\alpha(p\|q) = \text{KL}(p\|q).$$